

MCA II Semester
Artificial Intelligence – CS-411
Mid Term II

Time 1.30 hours

M.M. 20

Note:

- Read the question carefully before answering.
- Be precise in your answers.

1. Consider a sliding tile puzzle with 7 positions, with the initial state as follows

Initial State: [B C A _ D E F]

The objective is to arrange the tiles in sequence with a blank at the end. You can move a tile only into the adjacent blank space (left or right). Each move has a cost = 1.

Use the A* algorithm to find the optimal solution. Use an appropriate heuristic. All steps should be properly defined: state space, heuristic calculation, table for costs, open and closed lists.

Also perform greedy A* search up to 6 steps and see the effect on the result. 8

2. What is a consistent heuristic? Is consistent heuristic, always admissible? Justify your answer. 2
3. Give a PEAS description for an AI agent for an online recommendation system and describe its structure with your understanding and intuition of the domain. Given a choice to combine it with a goal-based or utility-based agent, what should be preferred? Consider the one that you think fits better and suggest how it can be incorporated in model based structure suggested by you. What are the pros and cons of using it? 8
4. What do you understand by a search contour? What can be its role in state space search? 2

MCA II Semester
Artificial Intelligence – CS-411
End Semester Examination

M.M. 50

Time 3 hours

Note :

Read question carefully before answering.

Be precise in your answers.

First question is compulsory

Answer any four questions from Question 2 to 6.

Q1. Answer all of the following questions

- a) What is the main difference a general computer program and an agent
 - b) What are three basic problems in hill climbing approach for search.
 - c) What is the problem with left recursion in PROLOG.
 - d) What is axiomatic system for deductive reasoning, what is its advantage over Natural Deduction System.
 - e) What is a frame in context of knowledge representation, how is it related to semantic network.
- 2X10

Q2(a). Given cities with symbolic names : (S,A,B,C,D,G) with possible paths connecting them. The cost of connecting path is given in form of weights. Find the shortest distance between S and G using A* algorithm. The heuristic function is given as straight line distance between source node to D

Edge	Cost
S → A	1
S → B	4
A → C	2
A → D	5
B → D	1
C → G	5
D → G	3

Source node	heuristics
S	6
A	5
B	2
C	4
D	1
G	0

Edges with Cost

Heuristic $g(n)$ = Straight line distance from G

Q 2(b). Consider tic tac toe game with players : X (MAX player) , O (Min player). Utility score of leaf nodes is as follows: +1 (X wins) -1 (O wins) 0 (Draw)

Current state of board is as follows :

	1	2	3
1	x	o	x
2	o	x	
3			o

Now it's MaX turn. Apply Min Max strategy to find best move for X out of three blank places. Is it possible that X can force a win at this point. (Hint : Make a complete tree considering all remaining actions, at leaf node assign utility values than perform min max search with alpha beta pruning)

$$5+5 = 10$$

Q3(a). Write the following statement in Predicate calculus

A computer is intelligent , if it can perform a task which if performed by human, requires intelligence.

3(b) Find which statements unify and give the result after unification. Identify those that do not unify, justify the reason.

- Unify(Knows(John,x),Knows(y,Mother(y)))
- Unify(Knows(John,x),Knows(x,Elizabeth))
- Unify(P(f(A),x),P(x,A))
- P(f(x,x),A),P(f(y,f(y,A)),A)

3© Given following sentences ,

Every student who studies Artificial Intelligence is intelligent.

Anyone who is intelligent can solve complex problems.

Rahul studies Artificial Intelligence.

Anyone who can solve complex problems gets good research opportunities.

- Represent them in Predicate calculus form.
- Using the above predicates, Use resolution refutation to prove that Rahul gives good research opportunities

$$2+4+6$$

Q4(a) Write a Prolog Program to find sum of N integers use the cut predicate, show how it improves accuracy of your program. The cut should not be a red cut, justify the same.

4(b) Given following natural language statements :

- Any student having good grades and has good programming skills is eligible for job.
- Anyone who is eligible for the job and clears interview gets the job

3. John is a student.
4. John has good grades.
5. John good programming skills.
6. John clears interview.

- (i) Write the corresponding Prolog facts and rules.
- (ii) Write the query to check whether John gets the job.
- (iii) Show the inference process.

4+6

Q5. Given following information :

Car is an automobile that is private carrier. An automobile is used for transportation. Automobile can be a public carrier or private carrier. Auto is also an automobile, it is public carrier. All automobiles have fixed seating capacity. A car has a brand such as : Maruti, tata. Alto is a car of brand Maruti. It is a family car. Tiago is a car of brand tata that is family car. Range Rover is a by Tata that is costly. An automobile may use petrol ,diesel or charged battery. A car may be a luxury car or family car. A luxury car is costly.

(a) Draw a semantic network from given statements . Clearly show standard links such as : isa, has part, instance. Show the properties using dashed edges. Give type of values that a property can take : numeric, string/user defined types .

(b) Give 4 inferences from the Semantic network and tell how they can be inferred. The inferences should be such that there are not present as natural language statements given to you. 8+2

Q6 (a) Differentiate between prior and posterior probability with example.

(b) How does Bayesian formula helps in dealing with uncertainty in knowledge representation.

(c) A dentist uses his nasty steel probe that catches in aching tooth of patient. Now considering two evidences Toothache and Catch (the events can be true or false). Use following probability distribution table to answer the question:

	Toothache		Not toothache	
	Catch	Not catch	Catch	Not catch
cavity	0.108	0.012	0.072	0.008
Not cavity	0.016	0.064	0.144	0.576

Using the table calculate following : .

A person has a toothache , what is the probability that he has a cavity

2+2+6